

On the Dynamics, Equilibria, and Economics of Gender Balance at Naturist Venues

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Abstract

This paper proposes a dynamical systems approach to modelling the demographic evolution of naturist venues. By quantifying gender-specific sensitivities to gender ratios, we derive an equation that governs the evolution of the ratio and that governs the system's trajectory. Stability analysis reveals a fundamental asymmetry: the "invisible hand" does not guarantee equilibrium, but rather admits a critical "tipping point" below which the venue collapses into a male-dominated state. We further analyse contentious "no single male" policy as a binding mathematical constraint and evaluate how it affects system dynamics and what side effects it has. Using KKT conditions, we derive the "shadow price" of female attendance, demonstrating that cross-subsidisation strategies (e.g. asymmetric pricing) are useful tools for both revenue and welfare maximisation. Finally, we introduce finite carrying capacity (supply-side constraint), concluding that sustainable venue management requires active flow control to maintain the demographic mixture within its stable basin of attraction. If this is infeasible, we propose alternative strategies in the conclusion.

On the Dynamics, Equilibria, and Economics of Gender Balance at Naturist Venues
Differential Model of Naturist Venue Demographics

To our knowledge, little effort has been put into quantifying or even characterising the implications and consequences of oft anecdotally discussed gender balance, or rather imbalance, observed in naturist venues. To this end, we propose a simple system of ordinary differential equations (ODEs) to model the evolution of demographics at such venues based on fixed coefficients quantifying potential participants' attitudes towards the ratio of genders.

Given f females and m males, over time t evolution is posited to obey

$$\begin{cases} \frac{df(t)}{dt} = \phi_1 \cdot \frac{f(t)}{m(t)} - \phi_2 \cdot \frac{m(t)}{f(t)} \\ \frac{dm(t)}{dt} = \mu_1 \cdot \frac{f(t)}{m(t)} - \mu_2 \cdot \frac{m(t)}{f(t)} \end{cases} \quad (1)$$

where ϕ_1 is females' preference for a higher female-to-male ratio f/m , ϕ_2 is females' dissatisfaction with a higher male-to-female ratio m/f , μ_1 is males' preference for a higher female-to-male ratio f/m , and μ_2 is males' dissatisfaction with a higher male-to-female ratio m/f . It is assumed, at least initially, that $\phi_1, \phi_2, \mu_1, \mu_2 > 0$ and that $\phi_1, \phi_2, \mu_1, \mu_2 \in \mathbb{R}$.

In some sense the system can be interpreted as an interplay between:

- (a) **“Comfort terms”** $f(t)/m(t)$, where a higher ratio of females to males generally increases the ‘comfort’ and “social safety” for everyone. For females, it reduces the feeling of being stared at or being the object of unwanted attention; for males, it validates the venue as a popular, balanced social environment; and
- (b) **“Pressure terms”** $m(t)/f(t)$, where a higher ratio of males to females creates a ‘skewed’ environment. In naturist venues, a high male-to-female ratio is often perceived as ‘creepy’ or ‘unbalanced’, leading to a decrease in the growth rate of both populations (though sensitivity can vary markedly, as discussed later).

It is assumed that *a priori* no inherent gender bias in favour or against practicing naturism exists but that observed rates of participation in public venues are driven by the

gender ratios themselves. We will find that females' displeasure at being in considerable minority and male relative indifference to high proportions of their own gender come to dominate the behaviour of the system. To state the obvious, as female proportion increases, both populations experience growth (or more precisely, an influx of participants), while as male proportion increases, both populations experience decline (or efflux).

The objective of such a toy model is to capture the essence of the phenomenon while remaining broadly tractable analytically and thus serving as a source of insight, rather than quantitative accuracy that would probably be better attained by other means. Nonetheless we shall find that such a simplistic model (further aided by finite capacity) has considerable descriptive power and captures many core features observed anecdotally in the field.

In the interests of brevity and clarity the dependence of f and m on t will be henceforth dropped from discussions of (1), leading to a restatement in condensed form:

$$\begin{cases} \dot{f} = \phi_1 \cdot \frac{f}{m} - \phi_2 \cdot \frac{m}{f} \\ \dot{m} = \mu_1 \cdot \frac{f}{m} - \mu_2 \cdot \frac{m}{f} \end{cases} \quad \text{with } \phi_i, \mu_i \in \mathbb{R}_{>0}. \quad (2)$$

To solve for the phase-space population trajectories, we first introduce the ratio function $r \doteq f/m > 0$ such that $f = xm$. Since the system is homogenous and of degree zero, the phase trajectories are determined by the relationship between f and m .

Using the quotient rule, we can take the derivative of r :

$$\dot{r} = \frac{\dot{f}m - f\dot{m}}{m^2} = \frac{\dot{f}}{m} - \frac{f}{m} \cdot \frac{\dot{m}}{m} \quad (3)$$

Substituting the expressions for $\dot{f}$ and $\dot{m}$ gives:

$$\dot{r} = \frac{1}{m} \left(\phi_1 - \frac{\phi_2}{r} \right) - \frac{r}{m} \left(\mu_1 r - \frac{\mu_2}{r} \right). \quad (4)$$

Factoring out $1/m$ and distributing r gives us

$$\dot{r} = \frac{1}{m} \left(-\mu_1 r^2 + \phi_1 r + \mu_2 - \frac{\phi_2}{r} \right) \quad (5)$$

To find trajectories in the (m, f) plane we eliminate time t by taking the ratio of the two original ODEs and then simplify by multiplying both numerator and denominator by the term f/m , and finally using $r = f/m$:

$$\frac{df}{dm} = \frac{\dot{f}}{\dot{m}} = \frac{\varphi_1 \frac{f}{m} - \varphi_2 \frac{m}{f}}{\mu_1 \frac{f}{m} - \mu_2 \frac{m}{f}} = \frac{\varphi_1 (\frac{f}{m})^2 - \varphi_2}{\mu_1 (\frac{f}{m})^2 - \mu_2} = \frac{\varphi_1 r^2 - \varphi_2}{\mu_1 r^2 - \mu_2} \quad (6)$$

Now we isolate the variables m and r to perform integration:

$$m \frac{dr}{dm} = \frac{\varphi_1 r^2 - \varphi_2}{\mu_1 r^2 - \mu_2} - r = \frac{\varphi_1 r^2 - \varphi_2 - \mu_1 r^3 + \mu_2 r}{\mu_1 r^2 - \mu_2}. \quad (7)$$

Separating the variables gives us:

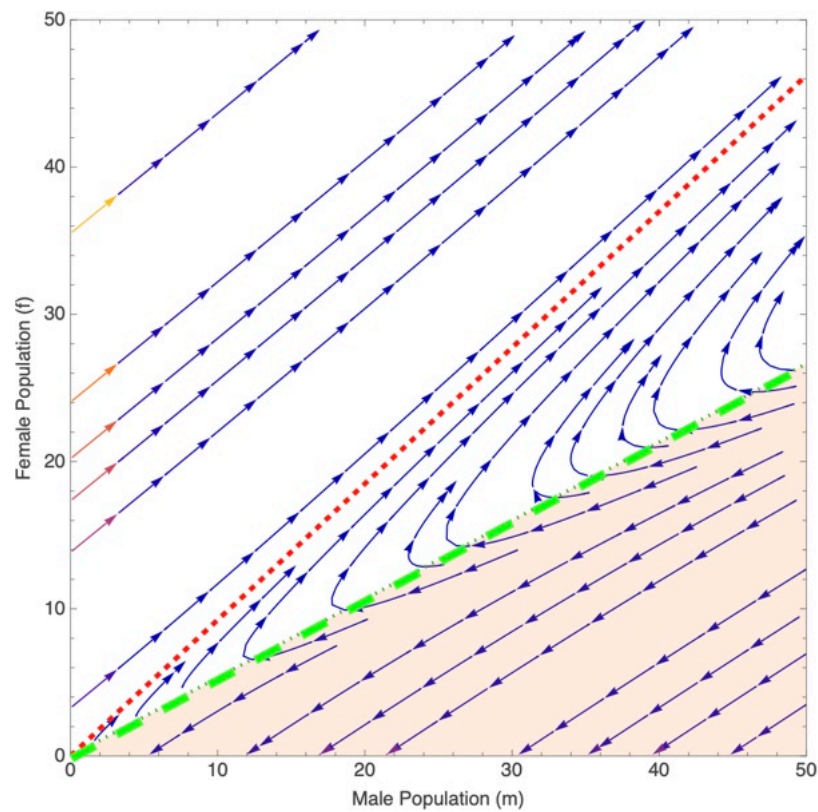
$$\frac{\mu_1 r^2 - \mu_2}{-\mu_1 r^3 + \varphi_1 r^2 + \mu_2 r - \varphi_2} dr = \frac{dm}{m}. \quad (8)$$

The integral on the left depends on the roots of the cubic polynomial $P(r) \doteq r(\mu_2 + \varphi_1 r - \mu_1 r^2 - \frac{\varphi_2}{r}) = -\mu_1 r^3 + \varphi_1 r^2 + \mu_2 r - \varphi_2$, and two cases are worth distinguishing:

- (a) **Fixed Rays** when $P(r) = 0$ has a positive real root r^* then $\frac{dr}{dm} = 0$ that represents a straight-line trajectory in phase-space where $f = r^* m$ that function as an attractor to which general curves tend to or repulsed by, but cannot cross; and
- (b) **General Curves** where $\dot{r} \neq 0$ the trajectories will curve towards or away from those fixed rays, defining the flow of the naturist venue's population levels.

In the sample diagram, the streamlines governing the system's behaviour are depicted in blue. The red dashed line is a fixed ray attractor, and in this example since it is below the 45° incline it shows circumstances where males (slightly) outweigh females. The green dashed line represents the isocline of zero female growth $\dot{f} = 0$ where the number of females joining exactly matches the number of females leaving. Below that, the shaded area represents a region where female population collapses to zero. A relatively high tolerance for predominantly male demographics ($\mu_2 \approx 0$) ensures the remaining population is all male.

Sample phase plot of male and female demographics at naturist venue



The one-line take-away is that in this system of ODEs modelling gender balance at naturist venues *equilibria are ratios of relative populations and not specific values.*

Stability of Phase-Space Solutions

Using the quotient rule on $r = f/m$ gives

$$\dot{r} = \frac{\dot{f}m - \dot{m}f}{m^2} = \frac{1}{m}(\dot{f} - r\dot{m}) \quad (9)$$

into which we can substitute our system equations, yielding

$$\dot{r} = \frac{1}{m} \left[\left(\varphi_1 r - \frac{\varphi_2}{r} \right) - r \left(\mu_1 r - \frac{\mu_2}{r} \right) \right]. \quad (10)$$

Combining denominators by multiplying within the bracket by r/r gives us the equation that governs the overall evolution of the gender ratio:

$$\dot{r} = \frac{1}{mr} (-\mu_1 r^3 + \varphi_1 r^2 + \mu_2 r - \varphi_2) = \frac{1}{mr} \cdot P(r) \quad (11)$$

Since the population m and the ratio r are strictly positive quantities, the sign of the rate of change of $\dot{r}$ depends entirely on the characteristic polynomial $P(r)$. The equilibrium points of the system are the roots r^* where

$$P(r^*) = 0. \quad (12)$$

Not all equilibria are stable, however. To determine long-term fate of the venue, we analyse the stability of these roots by examining the derivative of the growth-rate $d\dot{r}/dr$ at the equilibrium point itself. Using the chain rule on the equation that governs the overall evolution of the gender ratio (11):

$$\left. \frac{d\dot{r}}{dr} \right|_{r=r^*} = \frac{\frac{dP}{dr}(r^*)}{mr^*}. \quad (13)$$

Since the prefactor $1/(mr^*)$ is always positive, the stability is determined solely by the slope of the polynomial $P(r^*)$ as it crosses the axis:

If $P(r^*) < 0$ this r^* is a stable attractor because the polynomial slopes downwards towards this root, ensuring that any deviation will be self-correcting. This represents the venue's "social equilibrium" ratio. It is the value around which the venue's gender balance will naturally cluster.

If $P(r^*) > 0$ this r^* is an unstable repulsor because as the polynomial slopes upwards, any deviation will be self-reinforcing. This can be thought of as representing a “critical ratio”. If the ratio falls beneath his critical value, the self-reinforcing feedback loop takes hold and accelerates until the female proportion vanishes.

Since population m and ratio r are always positive or equal to zero, the sign of $\dot{r}$ is entirely determined by the sign of $P(r)$, and we have three distinct demographic behaviours:

If $P(r) > 0$ then the proportion of females increases;

If $P(r) = 0$ the system is in equilibrium and the relative proportions remain constant;

If $P(r) < 0$ the proportion of males increases.

Let us investigate the extrema of $P(r)$ in order to characterise its behaviour and thus perform boundary analysis keeping $m, f > 0$ because the model is undefined on the axes.

At $P(0) = -\varphi_2$. Since φ_2 (female exit rate) is positive, $P(0)$ must be negative. Demographically this implies that if the ratio is very low, the *change in the rate* $\dot{r}$ is negative, and the system will be driven towards total collapse ($r = 0$) and there is no mechanism of endogenous recovery from this final population of zero.

At $r \rightarrow \infty$ (infinite females) the leading term $-\mu_1 r^3$ dominates. Since μ_1 (male influx) is positive, $P(r) \rightarrow -\infty$. The demographic interpretation is that if the female population were to become overwhelmingly large somehow, a correspondingly oversized male influx will inevitably bring the ratio back down.

As the curve defined by $P(r)$ begins negative at $r = 0$ and goes to negative infinity at arbitrarily large r , there are only two scenarios for a venue:

No positive roots imply a “doomed resort” where the coefficients of $P(r)$ are such that its local maximum remains below zero, and thus $\dot{r}$ is always negative. Demographically this entails irrespective of how many females attend initially, the ratio will relentlessly decrease until the female population vanishes.

Two positive roots imply a “viable resort” where the attraction parameters (φ_1, μ_2) are strong enough relative to the “exit/entry” pressures to make the cubic curve bulge upwards and cross the axis twice at roots that we shall denote r_1 and r_2 . We can characterise these roots on the basis of the gradient of $P(r)$.

The lower root r_1 is a “viability threshold” where the curve makes its first crossing from negative to positive territory. Its slope is positive and this is an *unstable repulseor*, and the dynamics are as follows:

If the system begins slightly below this root ($r < r_1$), $P(r)$ is negative and $\dot{r} < 0$ so the system will be pushed down towards zero (collapse).

If the system begins slightly above this root ($r > r_1$), $P(r)$ is positive and $\dot{r} > 0$ so the system will be pushed upwards towards the second root. This is the “critical ratio” the venue must stay above in order to remain viable.

The upper root r_2 is a “natural equilibrium” where the curve would make a second crossing from positive territory back into negative values. The slope here is negative and it is a *stable attractor*: if the system is above this value, $\dot{r} < 0$ (the ratio decreases), and if the system is below this value, $\dot{r} > 0$ (the ratio increases). This represents the long-term destiny of the venue: as long as the ratio stays above this threshold, the demographics will naturally settle here.

Summarising the stability analysis, the general condition for a sustainable naturist venue in this model is that the polynomial $P(r)$ must have a local maximum greater than zero. We can find the local maximum by solving

$$P'(r) = -3\mu_1 r^2 + 2\varphi_1 r + \mu_2 = 0 \quad (14)$$

If the value of $P(r)$ at this maximum is positive, the venue will survive and maybe thrive. If it is negative, the resort is destined to fail.

The effects of male indifference to male predominance

Intuition guides into believing that males have a relatively high tolerance for predominantly male demographics $\mu_2 \approx 0$. We can harness this intuition and make males totally indifferent by setting $\mu_2 = 0$. This represents a resort where males do not depart even if females are vastly outnumbered. This changes the dynamics significantly: without the “pressure relief valve” of males departing, the venue is much more prone to collapse if the female population diminishes.

When $\mu_2 = 0$, the governing polynomial loses its linear term: $P(r) = -\mu_1 r^3 + \varphi_1 r^2 - \varphi_2$, creating a very specific “geometry of viability”. The function has a local maximum at $r_{max} = \frac{3\mu_1}{2\varphi_1}$, and in order for this venue to continue being viable, this peak must be positive ($P(r_{max}) > 0$), leading to the inequality

$$P(r_{max}) > 0 \iff \frac{4\phi_1^3}{27\mu_1^2} - \phi_2 > 0 \iff \phi_1^3 > \frac{27}{4}\mu_1^2\phi_2 \quad (15)$$

whose demographic interpretation is that the influx of females φ_1 relative to that of males μ_1 must be great enough to overcome the basic female efflux rate φ_2 . If this condition is satisfied, we have three roots for $P(r)$:

An unphysical negative root ρ_1 that can be ignored;

An unstable repulser threshold root ρ_2 beneath which ratio the venue is destined for collapse;

A stable attractor root ρ_3 at which the venue’s demographic ratio will naturally settle.

We are able to solve for the population trajectories in this scenario. We know that

$$\frac{dm}{m} = \frac{\mu_1 r^2}{-\mu_1(r - \rho_1)(r - \rho_2)(r - \rho_3)} dr. \quad (16)$$

Cancelling μ_1 and using partial fraction decomposition we have:

$$\frac{r^2}{(r - \rho_1)(r - \rho_2)(r - \rho_3)} = \frac{k_1}{r - \rho_1} + \frac{k_2}{r - \rho_2} + \frac{k_3}{r - \rho_3} \quad (17)$$

Integrating both sides gives us the exact shape of the curves in the phase portrait:

$$m(r) = C_0 \cdot |r - \rho_1|^{-k_1} \cdot |r - \rho_2|^{-k_2} \cdot |r - \rho_3|^{-k_3} \quad (18)$$

(where coefficients k_i are derived from roots such as $k_1 = \frac{\rho_1^2}{(\rho_1 - \rho_2)(\rho_1 - \rho_3)}$ & *cetera*).

We are remarkably also in a position to calculate $t(r)$: how long it takes a venue to reach a gender ratio (including the special case $t(0)$, the moment of collapse). We do this by substituting $m(r)$ into the male rate equation $\dot{r} = \frac{dm}{dt} = \mu_1 r$ to obtain:

$$t(r) = \frac{1}{\mu_1} \int_{r_0}^r \frac{1}{s \cdot m(s)} dm. \quad (19)$$

Recalling that $dm = m \frac{\mu_1 s^2}{P(s)} ds$ we can rewrite this as

$$t(r) = \int_{r_0}^r \frac{m(s) \cdot s}{P(s)} ds. \quad (20)$$

For illustrative purposes, consider the following “death spiral” scenario, wherein a venue starts with an initial ratio r_0 slightly below the threshold ρ_2 . $P(r)$ will be negative, so $\dot{r}$ will also be negative, and so the ratio begins to slide downwards. As it approaches $r = 0$, ever-smaller values of r cause the constant term $-\varphi_2$ to dominate the polynomial, and the rate of decline $\dot{r}$ becomes approximately proportional to $1/r$. A crash finally occurs because the integral converges to a finite value, meaning that the female proportion doesn’t just asymptotically fade away; it actually *hits zero in finite time*.

Regarding Closed-form Solutions in the Time Domain

We have seen that the trajectory of the population in the phase plane is determined by the integral relation between m and r :

$$\int \frac{dm}{m} = \int \frac{\mu_1 r^2 - \mu_2}{P(r)} dr \quad (21)$$

where $P(r) = -\mu_1 r^3 + \varphi_1 r^2 + \mu_2 r - \varphi_2$ is a full cubic polynomial. In order to solve this integral analytically, we must perform a partial fraction decomposition, which in turn requires factoring the denominator into the form $-\mu_1(r - \rho_1)(r - \rho_2)(r - \rho_3)$. Though soluble in general, the symbolic expressions for these roots involve nested cube roots of complex discriminants (Cardano's formulae). Substituting these large symbolic expressions into the exponents of a power-law solution renders the result opaque and algebraically intractable. Without first quantifying the coefficients, we cannot know if the roots are real, complex, distinct, or repeated.

This, unfortunately, is but the first and most easily dealt-with impediment. Even in fortunate or contrived circumstances where we find coefficients that yield three clean, real roots (e.g. $r \in \{1, 2, 3\}$), we are immediately met with a more formidable issue: inversion.

Let us assume, for the sake of argument, that we have found the trajectory $m(r)$. To find time evolution $r(t)$, we must solve the integral $t(r) = \int \frac{m(s) \cdot s}{P(s)} ds$ (17). Solutions to this integral will invariably take the form of a sum of logarithmic and perhaps polynomial terms:

$$t(r) = A \log_e |r - \rho_1| + B \log_e |r - \rho_2| + C \log_e |r - \rho_3| + D \quad (22)$$

To find explicit functions $f(t)$ and $m(t)$ we must invert this equation to find r as a function of t . Unfortunately an equation of the form $t = \log_e(r) + \log_e(r - k)$ is not generally expressible in elementary closed form; typically only implicit forms or special functions are required. The solution $r(t)$ notionally exists, but it will be a non-elementary implicit function (often involving the *Lambert W* function or worse for multiple logarithmic

terms). Therefore, no general closed-form formula for $f(t)$ can be given for this model. We are forced to rely on parametric descriptions of the form $(t(r), m(r), f(r))$.

While the general case is intractable, there are specific constraints on the coefficients that simplify the algebraic treatment enough to allow for explicit solutions.

If we assume males are indifferent about the gender ratio when entering, then $\mu_1 = 0$, and thus the term $-\mu_1 r^3$ vanishes from $P(r)$ which then becomes a quadratic equation $P_2(r) = \varphi_1 r^2 + \mu_2 r - \varphi_2$. Roots are then found easily via the quadratic formula or completion of the square. Integration leads to simpler inverse trigonometric or hyperbolic functions (such as $\arctan$ and $\operatorname{arctanh}$) which can sometimes be successfully inverted to find $r(t)$ explicitly.

If, on the other hand, we adopt a very ‘egalitarian’ standpoint where female and male sensitivities are identical ($\varphi_1 = \mu_1$ and $\varphi_2 = \mu_2$), the numerator and denominator in the phase trajectory will often share factors, which in turn means that ratio evolution $\dot{r}$ might simplify to a constant or a simple power, potentially allowing for trivial solutions such as linear growth $r(t) \propto t$. This might be plausible in certain scenarios, such as when a female and a male attend as a couple, or in cultures where Freikörperkultur (FKK) is unremarkable and broadly accepted.

We have already analysed the intuitively very plausible “indifferent males” ($\mu_2 = 0$) scenario, and though $P(r)$ remains a cubic, the loss of the linear term r often simplifies the root structure. If females were to also become indifferent to being in stark minority ($\varphi_2 = 0$) this would further simplify root structure, but observation and intuition suggest that φ_2 is large and significant, probably due to some combination of socially-enforced modesty ingrained into females and expectations of obnoxious or even predatory male behaviour. Solutions would be parametric as the inversion problem would still exist.

Forcing Viability: the “No Single Males” policy

We have so far discussed “free entry” models, where the influx of females and males were independent functions of the current ratio r : with natural female influx per unit time given by

$$J_f^{nat} = \varphi_1 \frac{f}{m} = \varphi_1 r, \text{ and natural male influx per unit time will be} \quad (23)$$

$$J_m^{nat} = \mu_1 \frac{f}{m} = \mu_1 r. \quad (24)$$

To formalise the “no single males” policy, we model the venue as an optimiser facing a binding constraint. The policy dictates that female entry must equal or exceed male entry $J_f \geq J_m$. In the strictest or most draconian interpretation, we force $\dot{m}_{in} \leq \dot{f}_{in}$, thus creating a system with a binding constraint.

In this scenario of constrained dynamics, evolution of the male population is now governed by the minimum of “natural influx” and “policy limit”:

$$\dot{m} = \min(\mu_1 r, \varphi_1 r) - \frac{\mu_2}{r} \quad (25)$$

Assuming the venue is popular enough that males want to enter at a higher rate than females do, (i.e., $\mu_1 > \varphi_1$ which is anecdotally not unreasonable and the root cause of the instability we modelled previously), the policy constraint becomes binding.

The system of ODEs that models the system collapse to:

$$\begin{cases} \dot{f} = \varphi_1 \cdot r - \frac{\varphi_2}{r} \\ \dot{m} = \varphi_1 \cdot r - \frac{\mu_2}{r} \end{cases} \quad \text{with } \varphi_i, \mu_i \in \mathbb{R}_{>0}. \quad (26)$$

(Note that in the expression for $\dot{m}$ μ_1 has been replaced by φ_1 .)

Recall $\dot{r} = \frac{1}{m}(\dot{f} - r\dot{m})$ from (9). Substituting in the constrained male rate (10) now

becomes:

$$\dot{r} = \frac{1}{m} \left[\left(\varphi_1 r - \frac{\varphi_2}{r} \right) - r \left(\varphi_1 r - \frac{\mu_2}{r} \right) \right] \quad (27)$$

which in turn simplifies to a modified version of (11), the equation that governs the overall gender ratio:

$$\dot{r} = \frac{1}{mr} (-\varphi_1 r^3 + \varphi_1 r^2 + \mu_2 r - \phi_2). \quad (28)$$

The original destabilising cubic term $-\mu_1 r^3$ has been replaced by the ‘policy’ term $-\varphi_1 r^3$. As we generally assume $\mu_1 > \varphi_1$ (males are more eager to frequent than females), replacing μ_1 with φ_1 significantly flattens the downward curve of the cubic polynomial $P_p(r)$. This pushes roots towards the left, effectively lowering the critical threshold required for viability. *Effectively, by tying male entry to female entry, policymakers artificially slow the rate of male growth to match the female ‘comfort’ growth rate and ensure the ratio is always viable.* The venue cannot become ‘oversaturated’ with males by the process of entry, only via differential exit (and as we have seen, μ_2 can be argued to be essentially zero).

Shadow Costs: Unlocking Value

In the unconstrained region (where the constraint is slack and not effective), the ‘cost’ of entry is zero (or the nominal access fee). However, when the constraint binds ($\dot{n}_{in} = \dot{f}_{in}$ despite $\mu_1 > \varphi_1$), we have *excess demand*.

There is now a difference between the number of males who wish to enter and the number who are granted access:

$$\Delta J_m = (\mu_1 - \varphi_1) \cdot r \quad (29)$$

The *shadow price* λ represents the marginal value of a female entrant, as every woman who enters ‘unlocks’ a corresponding slot for a male, the presence of a female guest acquires measure economic value equal to the profit that could be generated by that blocked male entrant.

We have already seen that unconstrained female demand flux is $J_f^{dem} = \varphi_1 \cdot r$ and unconstrained male demand flux is $J_m^{dem} = \mu_1 \cdot r$, however due to policy constraints aimed at curtailing instability actual male entry J_m cannot exceed actual female entry J_f :

$$J_m \leq J_f. \quad (30)$$

In this “excess male demand” scenario, the venue is turning away males that are willing to pay in order to be granted access, depressing its revenue.

Since J_m is capped by J_f , total admitted flux is determined solely by female demand J_f . For every single female who chooses to enter, one further male is admitted. We can optimise this in two distinct manners.

Scenario A: Maximisation of Revenue (Z)

Consider the venue to be a commercial operator seeking to maximise total revenue Z (a proxy for profit and economic utility). The objective function is

$$Z \doteq P_f \cdot J_f + P_m \cdot J_m \quad (31)$$

where P_i represents access fees paid by each demographic to the venue. Under the binding policy, the effective constrained revenue function becomes:

$$Z_{constrained} \doteq P_f \cdot J_f + P_m \cdot J_f = (P_f + P_m) \cdot J_f \quad (32)$$

Under this scenario, the marginal value of one additional female guest is no longer just her access fee P_f , but rather it has now become

$$\frac{\partial Z}{\partial f_{in}} = P_f + P_m \quad (33)$$

To rigorously ground this result, we invoke the Karush-Kuhn-Tucker (KKT) conditions (KARUSH, 1939; Kuhn and Tucker, 1951). Let $\lambda \geq 0$ be the Lagrange multiplier associated with the policy constraint $g(J_f, J_m) = J_m - J_f \leq 0$. The Lagrangian for the venue's maximisation problem is

$$\mathcal{L} \doteq P_m J_m + P_f \cdot J_f + \lambda (J_f - J_m). \quad (34)$$

The first-order stationary condition with respect to female entry is (in the minimisation convention)

$$\frac{\partial \mathcal{L}}{\partial J_f} = P_f + \lambda = 0 \quad (35)$$

or, more simply, the marginal value is lifted by λ . When the constraint is binding (male demand for access is exceeding the quota), the shadow price λ exactly equals the marginal revenue of excluded males (P_m), which should be subject to a highest-bidder auction in order to preferentially satisfy those who most desire to attend. Consequentially, the effective marginal value of a female guest becomes $V_f = P_f + \max(P_m)$. This multiplier λ

represents the formal economic justification for subsidies to support female attendance such as the schemes described in (Pollen, 2021, 2022): the subsidy creates a positive externality every time an additional female is enticed to attend, and to maximise total surplus, it must internalise this externality by subsidising her entry up to the value λ . The male premium pays for the female subsidy to relax the binding constraint.

Scenario B: Maximisation of Attendance (W)

Contrast this with a community-run venue seeking to maximise total participation & welfare W of its attendees, approximated here by total attendance

$$W \doteq J_f + J_m. \quad (36)$$

The optimisation problem is distinct, and the binding constraint yields the result

$$W_{constrained} \doteq 2J_f. \quad (37)$$

To maximise attendance and welfare, the venue must maximise female attendance J_f as each unlocks a corresponding male vacancy. Thus we arrive at a powerful conclusion: under the constraints of the “No Single Male” policy, the incentives of revenue-maximisation and welfare-maximisation are perfectly aligned. In both scenarios, the optimal strategy is invariant: the female guest is the limiting resource, and just as her preferences φ_1, φ_2 govern her attendance, her shadow price governs the potential scale of the venue’s overall attendance.

Trading Off: Marginal Rates of Substitution

Marginal Rate of Substitution (MRS) describes and quantifies the trade-off required to maintain a constant level of “social utility” (or growth stability) at the venue.

Let us define the utility of females for the venue U_f as equivalent to the female growth rate (a proxy for comfort/attractiveness):

$$U_f(m, f) \doteq \dot{f} = \varphi_1 \frac{f}{m} - \varphi_2 \frac{m}{f} \quad (38)$$

The MRS is thus the answer to the question: “if we add one male (increasing discomfort), how many females must we add to keep the overall comfort level unchanged?”

The slope of the iso-comfort curve $dU_f = 0$ is given by:

$$\frac{\partial U_f}{\partial f} + \frac{\partial U_f}{\partial m} = 0. \quad (39)$$

Taking the partial derivatives gives:

$$\frac{\partial U_f}{\partial f} = \frac{\varphi_1}{m} + \frac{\varphi_2 m}{f^2}, \quad \frac{\partial U_f}{\partial m} = \frac{\varphi_1 f}{m^2} - \frac{\varphi_2}{f} \quad (40)$$

Solving for the gradient $\frac{df}{dm}$ gives:

$$MRS_{m,f} = -\frac{df}{dm} = \frac{-\left(-\frac{\varphi_1 f}{m^2} - \frac{\varphi_2}{f}\right)}{\frac{\varphi_1}{m} + \frac{\varphi_2 m}{f^2}}. \quad (41)$$

Multiplying both numerator and denominator by m gives:

$$MRS_{m,f} = \frac{\varphi_1 r + \varphi_2 / r}{\varphi_1 + \varphi_2 / r^2} = r \cdot \left(\frac{\varphi_1 r^2 + \varphi_2}{\varphi_1 r^2 + \varphi_2} \right) = r, \text{ and thus } MRS = \frac{f}{m}. \quad (42)$$

In demographic terms, this means that in order to maintain the exact same level of comfort, the substitution rate is simply the current ratio. This both confirms and follows from isoclines (lines of constant growth) being rays from the origin. To keep the comfort constant, the venue must expand population linearly along the ray.

Full House: Finite Carrying Capacity

So far, our model has assumed unconstrained growth. However, as every physical venue has a finite carrying capacity K (determined by locker counts, sun-beds, limited accommodation or camping pitches, pool safety limits, or fire codes). Unlike biological and demographical systems where growth slows due to resource scarcity (the logistic damper of Verhulst), a venue usually hits a hard constraint when whatever limiting resource is fully subscribed. Furthermore, as potential attendees are neither being born nor dying according to the abundance of resources, but simply turned away when K is reached, it seems preferable to model a “turnstile model”¹: not a damping term on the parameters, but a boundary condition on the domain. The system evolves according to its standard equations until the total population $f + m = N = K$. We can formalise this as a constraint:

$$f + m \leq K \tag{43}$$

1. There is an additional issue of information asymmetry to be taken into account: naturist venues, by their very nature, tend to be private and not easily viewable from outside. This entails that prospective entrants presenting themselves at the gate will be unaware of the current level of attendance or crowding inside the venue’s borders.

Dynamics at Capacity: the “Churn Mechanism”

Once the venue is full and the constraint

$$f + m = K \quad (44)$$

holds, dynamics switch from growth to replacement. We must also be careful to maintain the demographic balance between f and m in order to avoid a “death spiral” by ensuring demographics must settle on the stable attractor ray

$$f = r^* m \quad (45)$$

We are going to solve for the point that satisfies both conditions simultaneously. Substituting the stable ratio constraint (45) into the capacity constraint (44) results in:

$$m + r^* m = K \quad (46)$$

whereupon we can factor out m to obtain

$$m(1 + r^*) = K. \quad (47)$$

Now we can solve for the male equilibrium population m^* :

$$m^* = \frac{K}{1 + r^*} \quad (48)$$

and then substitute back into (44) to find the female equilibrium

$$f^* = \frac{K r^*}{1 + r^*} \quad (49)$$

This point (f^*, m^*) represents the only sustainable “full occupancy” scenario. If a venue reaches capacity at a different ratio, the system is in one of two conditions:

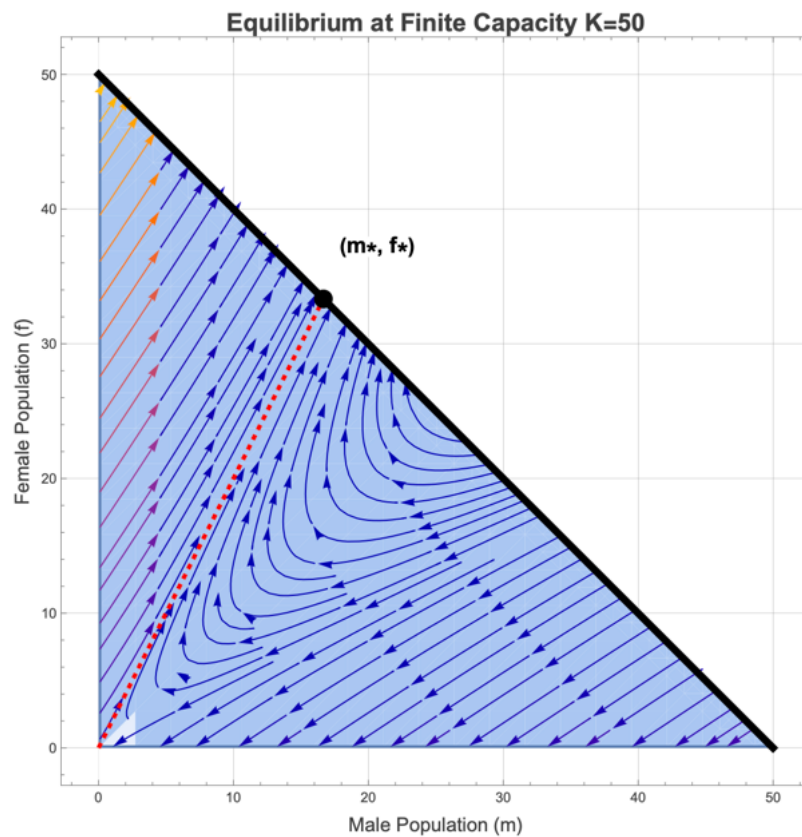
If $r > r^*$ the venue has a female-heavy demographic. The venue is stable and saturated but admissible males cannot gain entry. If the venue differentiates price by gender and subsidises female participation, it is leaving revenue on the table.

If $r < r^*$ the venue has a female-heavy demographic and is in the unstable ‘repulsor’ zone. In a regime of mere replacement, the system cannot auto-correct: suppose the venue hits

capacity with a male-dominated ratio $r < r^*$. Males, having a lower sensitivity to gender imbalance $\mu_2 \approx 0$ are unlikely to leave. However, females, feeling crowded out and in discomfort, will exit ($\dot{f}_{out} > 0$). These female exits create new availability for new entrants which by definition will have to be females, because at capacity any vacancy created by a female is going to be reserved for the entry of a new female guest. However, replacing dissatisfied females with other dissatisfied females does not restore equilibrium but maintains the $r < r^*$ regime.

In order to avoid this, the venue must most likely run at $N < K$ (below capacity), sacrificing immediate revenue in order to prevent irrecoverable female population collapse.

Sample phase plot of male and female demographics at naturist venue with finite carrying capacity = 50



Conclusion

This paper presented a dynamical systems approach to modelling the demographics of naturist venues. By quantifying gender-specific preferences for, and sensitivities to, population ratios, we derived an equation that overall evolution of the gender ratio (11) that governs evolution of the venue's social balance.

Our analysis yields three principal theoretical conclusions:

1. Inherent Instability

The proverbial “invisible hand” does not guarantee a balanced gender ratio and to some extent can punch a venue in the groin. We identified the existence of a critical unstable root below which the venue enters a “death spiral” of male-dominated stagnation. This indicates that passive management is insufficient because a venue cannot organically “grow its way out” of a bad initial ratio if our intuitions about coefficients are correct ($\varphi_2 \gg 0, \mu_2 \approx 0$).

2. Shadow Price of Balance

By modelling the “no single males” policy as a binding constraint, we derived KKT conditions for the system. We demonstrated that under this constraint, the shadow price of a female entrant includes the opportunity cost of the male entrants she enables. This provides a rigorous economic justification for cross-subsidisation (e.g., “ladies free” pricing), showing that revenue maximisation (Z) (as a proxy for economic utility) and community participation maximisation (W) effectively converge on the same strategy: prioritising female attendance and participation.

3. The Determinism of Capacity

We showed that when a venue reaches its physical carrying capacity K , the final equilibrium (f^*, m^*) is strictly determined by the intersection of the stable gender ratio and the carrying capacity border.

This model suggests that a naturist venue is not merely a static facility but a complex social reactor. It suggests that care must be exercised when priming a naturist venue with its first participants, and that seeding the initial population with a selection of females who are largely indifferent to being outnumbered by males $\varphi_2 \approx 0$ could go a long way towards avoiding the scenarios of collapse the model predicts when imbalance sets in beneath the critical ratio that would ordinarily trigger the “death spiral” process.

In closing, some remarks about the “no single male” policy are in order, as it is notoriously contentious. Given the model and the assumptions about coefficients, it is shown to be an effective policy to prevent collapse of the female population from which there is no endogenous return. Whether a venue chooses to adopt it or not is largely down to their values and whether the preferences of their prospective female guests (as described in aggregate by their coefficients) make the prospect of female attendee collapse a realistic scenario which warrants guarding against. As noted above, even a small “hard-core” kernel of females unconcerned by males significantly outnumbering them can provide the bedrock that in turn could allow slightly more sensitive, but nonetheless less sensitive than average, females to participate. While the mathematical model emphasises the primacy of female preferences and behaviours, it is likely that male behaviour and attitudes are a determining factor in how at ease females feel at venues and therefore how inclined they are to stay as opposed to depart when gender ratios are unbalanced in their disfavour. When in heterosexual couples, both partners empirically display a general indifference to gender imbalance ($\varphi_2 \approx \mu_2 \approx 0$), presumably because the female feels protected. We must also keep in mind that coefficients are aggregate values, and that if there are enough females with $\varphi_2 \approx 0$, the community

average can decrease to the point that the unstable root disappears entirely, making the “death spiral” scenario impossible.

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Appendix

A key question for empirical verification is whether the theoretical coefficients $(\varphi_1, \varphi_2, \mu_1, \mu_2)$ can be recovered from real-world data collected in the field. While a single observation of a venue's gender balance (observing only r or more likely r^* if in equilibrium) presents an identification problem because an infinite set of parameter combinations can yield the same root, a time-series of attendance provides sufficient variation to estimate coefficients. Recall original model formulated in (1) and (2).

For a discrete time-series (taken at hourly or daily intervals) we approximate the instantaneous rate of a gender $\dot{g}_{f \vee m}$ as the finite difference

$$\Delta g_{t,f \vee g} = g_{t+1,f \vee g} - g_{t,f \vee g}, \quad (1)$$

that can be rearranged into a standard linear regression model of the form

$$Y = \beta_1 X_1 + \beta_2 + \epsilon. \quad (2)$$

Estimating female coefficients

Let the dependent variable be the net change in females:

$$Y_{f,t} = f_{t+1} - f_t \quad (3)$$

and let the independent variables be the current ratios:

$$X_{1,t} = \frac{f_t}{m_t}, X_{2,t} = \frac{m_t}{f_t}. \quad (4)$$

We estimate the regression

$$Y_{f,t} = \varphi_1 X_{1,t} - \varphi_2 X_{2,t} + \epsilon, \quad (5)$$

where the regression coefficient for $X_{1,t}$ gives the estimate for φ_1 and the coefficient for $X_{2,t}$ gives the estimate for $-\varphi_2$.

Estimating male coefficients

Similarly, let

$$Y_{m,t} = m_{t+1} - m_t. \quad (6)$$

We estimate

$$Y_{m,t} = \mu_1 X_{1,t} - \mu_2 X_{2,t} + \epsilon. \quad (7)$$

Data Requirements

To obtain statistically significant estimates, the data must contain **variance in the ratio r** . If, on the other hand, a venue is perfectly managed and always sits at equilibrium $r^* = f^*/r^*$, the regressors X_1 and X_2 will be unchanging constants. Similarly if observations are made in situations where there is no influx or efflux, such as on a nude cruise. In these cases regression will suffer from perfect collinearity, rendering the coefficients unidentified.

Valuable data is generated when the system is out of equilibrium. An “event study” approach analysing days when the ratio is eternally perturbed (e.g., a batchelor or bachelorette party arrives or departs) provides a useful exogenous shock to trace out response curves (X values) and the system’s correction (Y values).

Implied Structural Tests

Once the coefficients are estimated from the data, the theoretical “Survival Inequality” derived from this paper can be empirically tested. Though as initially stated this is only intended to describe a toy model amenable to analysis, it would be nonetheless interesting to ascertain to what degree it qualitatively and quantitatively describes actual observations in the field. If viable venues’ guests’ estimated parameters are found to be broadly compatible with those prescribed by the characteristic polynomial $P(r)$, this would justify it as being a potential object of study and social engineering to ensure robustness.